Galactose metabolism and health

Ana I. Coelho[a], *Gerard T. Berry*[b,*], *and M. Estela Rubio-Gozalbo*[a,*]

Purpose of review
Galactose – a key source of energy and a crucial structural element in complex molecules – is particularly important for early human development. However, galactose metabolism might be important not only for fetal and neonatal development but also for adulthood, as evidenced by the inherited disorders of galactose metabolism. The purpose of this review is to summarize the current evidence of galactose metabolism in health and disease.

Recent findings
The biological importance of galactose goes beyond its importance as a nutrient and a metabolite. Galactose has been selected by evolutionary pressure to exert also a crucial structural role in macromolecules. Additionally, galactose has recently been reported as beneficial in a number of diseases, particularly in those affecting the brain.

Summary
Galactose is crucial for human metabolism, with an established role in energy delivery and galactosylation of complex molecules, and evidence for other roles is emerging.

Keywords
energy, galactose, glycosylation, metabolism

INTRODUCTION

Galactose was first identified in milk by Louis Pasteur in 1856, who denominated it as 'lactose'. Only later, it was named 'galactose' from the Greek word 'galakt', which means 'milk'.

Galactose is a natural aldohexose that, like most sugars, occurs more frequently in nature in its D-configuration. D-galactose is ubiquitous in bacteria, plants, and animals [1]. It is available as free and bound galactose. The bound form comprises complex carbohydrates, for example, oligosaccharides and polysaccharides, glycoproteins, and glycolipids. Along with glucose, galactose forms the disaccharide lactose – a sugar present in most animal milks and a key source of energy in infants.

The biological importance of galactose, however, goes beyond its importance as a nutrient and a metabolite. Galactose appears to have been selected by evolutionary pressure to also exert a crucial structural role. Indeed, despite the fact that it differs from glucose in the configuration of the hydroxyl group at the carbon-4 position, galactose has a myriad of specific functional and structural roles in living organisms that cannot be exerted by glucose.

We review the importance of galactose for human health, with a special emphasis on its functional and structural roles.

GALACTOSE IN MILK

Milk composition varies greatly across mammalian species, as an adaptation to environment and nutrient requirements of the different mammalian infants [2,3]. Only the sea lion and the marsupials – largely inhabitants of the Pacific Ocean, are known to have milk without lactose, and still their first pouch milk is a trisaccharide of galactose [4].

In comparison with milk from other species, human milk is considered unique in terms of its sugar content. It contains 55–70 g/l of lactose and 5.0–8.0 g/l of complex oligosaccharides. There are over 100 different oligosaccharide structures in

[a]Department of Pediatrics and Laboratory Genetic Metabolic Diseases, Maastricht University Medical Center, Maastricht, The Netherlands and [b]The Manton Center for Orphan Disease Research, Division of Genetics and Genomics, Boston Children's Hospital, Harvard Medical School, Boston, Massachusetts, USA

Correspondence to Gerard T. Berry, The Manton Center for Orphan Disease Research, Division of Genetics and Genomics, Boston Children's Hospital, Harvard Medical School, 3 Blackfan Circle, Center for Life Science Building, Suite 14070, Boston, MA 02115, USA. Tel: +1 617 355 4316; e-mail: Gerard.Berry@childrens.harvard.edu

*Gerard T. Berry and M. Estela Rubio-Gozalbo contributed equally to the writing of this article.

Curr Opin Clin Nutr Metab Care 2015, 18:422–427
DOI:10.1097/MCO.0000000000000189

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

KEY POINTS

- Galactose is ubiquitous in all living organisms.
- Galactose exerts a myriad of functional and structural roles.
- Disorders of galactose metabolism reinforce the notion that galactose is crucial for the human body.

human milk, of which galactose is a main component [5]. The great majority carries lactose at their reducing end, and the core molecule is characterized by repetitive attachment of galactose and N-acetylglucosamine (GlcNAc) in β-glycosidic linkage to lactose [3,6].

Milk sugars ensure that galactose levels do not become limiting during postnatal early development, due to its crucial structural function, for example, for the brain [3,7▪]. Additionally, the sugars present in milk are a determinant factor in neonatal host defense and inflammatory processes due to their prebiotic effect, and are an important source of energy in infants, in whom lactose by itself accounts for 40% of calories and for 90% of dietary sugar [3,5,7▪].

GALACTOSE METABOLISM

Lactose is a disaccharide of galactose and glucose that is hydrolyzed by the disaccharidase lactase (β-galactosidase) into its constituent monosaccharides in the brush border membranes of the enterocytes. Galactose and glucose can then be absorbed across the small intestine by the mature enterocytes at the tips of the villi [8▪▪].

Galactose and glucose are actively transported across the brush border membrane by the Na^+/glucose co-transporter or symporter sodium/glucose co-transporter 1 (SGLT1) (K_M galactose ≈K_M glucose = 0.1–0.6 mmol/l; K_M is essentially identical for both sugars), in contrast to fructose that is absorbed passively by the facilitated transporter glucose transporter 5 (GLUT5). Two sodium ions first bind to the outer face of the transporter, causing a conformational change that allows galactose (or glucose) to bind. At the inner face, galactose (or glucose) is first released into the cytoplasm, followed by the sodium ions, thus regenerating SGLT1 for further Na^+/sugar transport. The Na^+ electrochemical potential gradient is maintained by the basolateral $3Na^+/2K^+$-ATPase, which pumps the co-transported sodium ions out across the basolateral membrane [8▪▪]. All three sugars are transported across the basolateral membrane by a passive process by GLUT2 (K_M glucose = 50 mmol/l; K_M galactose = K_M fructose = 66 mmol/l) [8▪▪,9]. The presence of high concentrations of sugar in the intestinal lumen triggers the translocation of GLUT2 to the apical membrane [8▪▪,10].

After efflux from the enterocyte, galactose enters the portal vein and is transported into the liver, where it is internalized by GLUT2 (K_M galactose ≈92 mmol/l) [10,11]. Due to the low affinity and high capacity of GLUT2 and immediate phosphorylation to galactose-1-phosphate (Gal-1-P), a major part (88% on average) of the ingested galactose is retained in the liver. The remaining amount is directly transported into the other organs and tissues, such as the brain, where it can be used to produce amino acids, and the lactating mammary gland, where it is used to produce lactose [3,10,12].

The distinctive hepatic capacity to retain and eliminate galactose can be used as a clinical measure of metabolic liver function [13,14]. Thus, liver failure results in galactose intolerance. Alcohol also results in an abnormal galactose tolerance test in humans.

THE LELOIR PATHWAY

Galactose exists in two predominant forms in aqueous solution: the α and β-pyranose structures, which differ in the configuration of the hydroxyl group at carbon-1 of the ring. Upon its release from lactose, galactose is in its β-conformation. β-Galactose is then converted into its α-anomer by galactose mutarotase (GALM, EC 5.1.3.3) [15].

Galactose mutarotase is an aldose 1-epimerase that catalyzes the interconversion of α and β-hexose sugars. The conversion of galactose into its α-conformation allows it to enter the Leloir pathway – the main pathway of galactose metabolism. This pathway metabolizes α-galactose through the subsequent action of three enzymes: galactokinase (GALK), which converts galactose into Gal-1-P; Gal-1-P uridylyltransferase (GALT), which converts Gal-1-P and uridine diphosphate-glucose (UDP-glc) into glucose-1-phosphate (Glc-1-P) and uridine diphosphate-galactose (UDP-gal); and UDP-galactose 4'-epimerase (GALE), which is responsible for the interconversion of UDP-glc and UDP-gal (Fig. 1).

Galactokinase (EC 2.7.1.6) belongs to the GHMP family (GALK, homoserine kinase, mevalonate kinase, and phosphomevalonate kinase) – a family of small-molecule kinases. It specifically phosphorylates α-galactose, using ATP as a cofactor (K_M galactose = 319 μmol/l and K_M ATP = 20.9 μmol/l). The reaction proceeds through the formation of an ordered ternary complex, in which ATP binds first, followed by galactose. Gal-1-P – the product of the reaction – is an inhibitor of GALK [15–17].

Galactose-1-phosphate uridylyltransferase is a member of the transferase branch of the histidine

1363-1950 Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

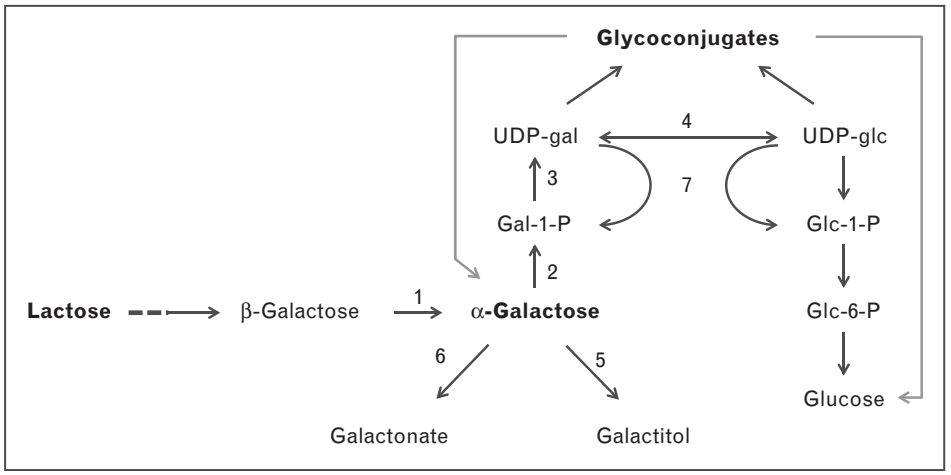


FIGURE 1. Overview of galactose metabolism: before entering the Leloir pathway, β-galactose must be first converted its α anomer by galactose mutarotase (GALM; enzyme 1). In the Leloir pathway, α-galactose is first phosphorylated into galactose-1-phosphate (Gal-1-P) by galactokinase (GALK; enzyme 2). Galactose-1-phosphate uridylyltransferase (GALT, enzyme 3) then transfers a uridine monophosphate (UMP) group from UDP-glucose (UDP-glc) to Gal-1-P, forming glucose-1-phosphate (Glc-1-P) and UDP-galactose (UDP-gal). In the third step of the Leloir pathway, UDP-galactose 4′-epimerase (GALE; enzyme 4) catalyzes the interconversion of UDP-gal and UDP-glc. Both UDP-gal and UDP-glc are sugar donors of the glycosylation reactions, important for the production of glycoconjugates. The Glc-1-P formed leads to the production of glucose-6-phosphate (Glc-6-P) and glucose, thereby leading to the production of energy. In an alternative to the Leloir pathway, three accessory pathways exist: the conversion of galactose into galactitol by aldose reductase (enzyme 5), the conversion of galactose into galactonate by galactose dehydrogenase (enzyme 6), and the conversion of Gal-1-P into UDP-gal by UDP-glucose/galactose pyrophosphorylase (UGP; enzyme 7).

triad (HIT) family of enzymes, displaying a ping-pong kinetics and a double displacement mechanism, in which the nucleophilic histidine at residue 186 is transiently nucleotidylated. First, UDP-glc enters the active site, binds covalently to His186, and then Glc-1-P is released. After that, Gal-1-P enters the active site, and the enzyme-bound uridine monophosphate (UMP) is transferred to Gal-1-P, forming and releasing UDP-gal [18].

Uridine-galactose 4′-epimerase (EC 5.1.3.2) is a member of the short-chain dehydrogenases/reductases family, and is responsible for the interconversion of UDP-gal to UDP-glc, as well as of UDP-N-acetylgalactosamine to UDP-N-acetylglucosamine, in mammals. It contains a bound NAD^+ as an essential cofactor, which is thought to transiently oxidize the sugar moiety at carbon-4 and then re-reduces it in a nonstereospecific manner, permitting inversion of configuration [19].

The net result of the Leloir pathway is the production of Glc-1-P, which can be further metabolized to release energy following conversion to glucose-6-phosphate, and the production of UDP-gal, the galactose donor in the glycosylation reactions.

Under physiological conditions, galactose is rapidly converted to glucose by the Leloir pathway. The body capacity of disposing galactose is indeed highly efficient: the glucose pools exhibit a 50% rise within 30 min after an oral or intravenous galactose load. Interestingly, when compared to glucose and fructose, galactose is preferentially incorporated into the liver glycogen, rather than to follow the oxidation route [20].

The liver appears to be the most important organ involved in the disposition of galactose. Nevertheless, the enzymes involved in the metabolism of galactose have been identified in several cells and tissues.

ACCESSORY PATHWAYS OF GALACTOSE METABOLISM

Although the Leloir pathway is the main pathway of galactose metabolism, alternative pathways also exist. These include galactose reduction, catalyzed by the enzyme aldose reductase (EC 1.1.1.21) and leading to the production of galactitol, which cannot be further metabolized, and, due to its poor diffusivity, accumulates in the cells; galactose oxidation, catalyzed by galactose dehydrogenase (EC 1.1.1.48) and leading to the production of galactonate, which can be directly excreted in urine or further metabolized through the pentose phosphate pathway; and the pyrophosphorylase pathway, which converts galactose to UDP-glucose by the sequential activities of GALK, UDP-glucose/galactose pyrophosphorylase (UGP, EC 2.7.7.10), and GALE [21].

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

BIOLOGICAL IMPORTANCE OF THE LELOIR PATHWAY

The Leloir pathway is strongly conserved in nature, from bacteria to yeast, and humans, highlighting the significance of galactose for living organisms. Its importance for the human development is further emphasized by the description of GALK and GALT detection in fetal tissue already in the 10th week of pregnancy. Furthermore, their specific activities in the fetus show an increase with gestational age, reaching a maximum around birth, with subsequent postnatal decrease [15].

Developmental modulation of the hepatic Leloir enzymes in the fetus, newborn, and pregnant rats has been the focus of intense research in the 70s and 80s. Each enzyme shows a specific developmental pattern. In contrast to GALK, which remains relatively constant, doubling only at term, GALT and GALE increase during fetal development, achieving a plateau that persists until birth. In newborn rats, GALK and GALT-specific activities increased until approximately day 8, at which point both show an intense peak, and then diminish to adult levels by day 21, whereas that of GALE remains elevated until day 15, followed by a decrease to adult levels by day 21. These metabolic studies underline the increased ability of the young liver to metabolize galactose, in line with the high enzyme activities observed during the suckling period when galactose is pivotal for development.

Additionally, studies on rat nervous system during late embryonic and early postnatal period revealed that GALT expression was exceptionally high in myelinating oligodendrocytes and Schwann cells after birth. The peak of expression correlated with the period of myelinogenesis, perhaps, in turn, related to the high content of galactocerebrosides in myelin. In late gestation, liver of pregnant rats shows an increase in all the three enzymes, particularly for GALT, with subsequent decrease following parturition. Nevertheless, GALT remains higher, whereas GALK and GALE revert to levels slightly lower than that of nonpregnant female rats [15].

The Leloir pathway leads to the production of UDP-gal – the galactose donor of the glycosylation reactions – thus being important for the glycosylation process.

Uridine diphosphate-sugars are the obligate sugar donors in the synthesis of complex glycoconjugates. They are produced in the cytoplasm, and then they are actively transported into the endoplasmic reticulum and the Golgi lumens by the nucleotide sugar transporters (NSTs), which are used as substrates by the glycosyltransferases [22,23▪].

The third and last enzyme of the Leloir pathway is responsible for the interconversion of UDP-gal to UDP-glc, and UDP-N-acetylgalactosamine to UDP-N-acetylglucosamine. GALE oversees the ratios of key substrate pools, due to their importance for cellular biology in the glycosylation process. Indeed, glycosylation plays a key role in a myriad of fundamental cellular processes, including growth, differentiation, migration, morphogenesis, and cell-to-cell interactions [24▪▪].

The Leloir pathway, apart from its importance for energy production, is crucial for the glycosylation of complex molecules, such as myelin, in which galactocerebroside is the predominant glycolipid. In fact, galactose was initially also called cerebrose, due to its functional significance for the brain.

In light of the galactose importance for development, it is therefore not surprising that the human body has the ability to synthesize de-novo galactose. This phenomenon was first recognized by Gitzelmann [21], who proposed that a UDP-galactose pyrophosphorylase reaction was the biochemical basis for the endogenous synthesis of galactose. Actually, the major source of de-novo galactose is more likely the Golgi-dependent synthesis and turnover of galactose-containing macromolecules, such as glycoproteins and glycolipids [21].

Endogenous production rates are higher in infants and young children compared to adolescents and adults. In fact, the rate of endogenous galactose production per body weight, when plotted against age, suggests a typical pediatric developmental growth curve. Therefore, the rate of endogenous synthesis may be much higher in prenatal fetal tissues than after birth [25], reinforcing galactose importance for early development. When an infant and child's growth velocity is plotted against age, the curve is exponential in nature, with the growth rate being maximal at birth.

REGULATION OF THE LELOIR ENZYMES

The human body is capable of quickly metabolizing galactose through the Leloir pathway. The removal mechanism of galactose from the blood is saturated at 50 mg/dl, believed to be secondary to the limited ability of GALK to phosphorylate the sugar.

After entering the cell, galactose is quickly phosphorylated by GALK, thereby entrapping it inside the cell. GALK is therefore considered the rate-limiting enzyme of the Leloir pathway. Studies on the specific activity of each of the enzymes on the liver of adult rats revealed, however, that GALE is the enzyme with the lower specific activity, which suggests that GALE is actually the rate-limiting enzyme of the Leloir pathway. This observation might be related with the biological importance of GALE in keeping the nucleotide sugars within well defined ratios.

1363-1950 Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

GALACTOSE IN DISEASE

Disorders of galactose metabolism have provided additional evidence of the importance of this sugar to the human body.

Deficient activity of any of the enzymes of the Leloir pathway leads to galactosemia. The most common and well known type of galactosemia is classic galactosemia (OMIM #230400), which is caused by a severe deficiency in GALT activity.

Classic galactosemia is a potentially lethal disorder in the neonatal period, with multiorgan involvement. Infants accumulate galactose and Gal-1-P to milimolar concentrations in target tissues, such as the liver and the brain [25]. Upon implementation of a galactose-restricted diet, the neonatal symptoms are quickly resolved, and both the metabolites decrease markedly. However, even patients diagnosed soon after birth and put immediately on a galactose-restricted diet develop burdensome complications on the long run, such as cognitive and fertility impairments, and Gal-1-P levels always remain elevated when compared to healthy infants [21,26].

Several pathogenic agents have been described in classic galactosemia. Metabolite toxicity by Gal-1-P accumulation and UDP-hexoses deficiency has been extensively pointed out as a major pathogenic factor [27–29]. In addition, a deficiency of *myo*-inositol has also been reported [25]. More recently, GALT misfolding and aggregation were implicated in the molecular basis of classic galactosemia [30,31]. The exact mechanism(s) by which these patients develop the long-term complications is, at present, not fully understood [32]. Nevertheless, classic galactosemia's severe outcome does reinforce the notion that galactose is indeed a pivotal sugar for human development.

GALACTOSE IN THERAPEUTICS

In addition to its broad role in human physiology, galactose has been recently reported as beneficial in a number of diseases, particularly in those affecting the brain functions [12,33▪]. Additionally, fibroblasts of patients with phosphoglucomutase 1 deficiency showed an improved glycosylation upon galactose supplementation, and the patients who received dietary supplementation showed some potentially important clinical changes [34▪▪].

Additional studies are warranted to further understand galactose mechanism(s) of action and to establish beyond doubt its therapeutic role.

CONCLUSION

Despite having been identified in 1856, at present, galactose is still a point of amazement.

Galactose is crucial for human metabolism, with an established role in energy delivery and galactosylation of complex molecules. Its main metabolic pathway is highly conserved in nature, being present in all living organisms. In humans, galactose is particularly important in early development. Genetic disorders that impair its metabolism inevitably cause disease, drawing attention to its key role.

Acknowledgements

None.

Financial support and sponsorship

None.

Conflicts of interest

There are no conflicts of interest.

REFERENCES AND RECOMMENDED READING

Papers of particular interest, published within the annual period of review, have been highlighted as:

- ▪ of special interest
- ▪▪ of outstanding interest

1. Bell D, editor. Natural monosaccharides and oligosaccharides: their structures and occurrence. In Florkin M, editor. Comparative biochemistry: a comprehensive treatise. Vol. 3. Elsevier; 2012. pp. 287–354.
2. Georgi G, Bartke N, Wiens F, Stahl B. Functional glycans and glycoconjugates in human milk. Am J Clin Nutr 2013; 98:578S–585S.
3. Chichlowski M, German JB, Lebrilla CB, Mills DA. The influence of milk oligosaccharides on microbiota of infants: opportunities for formulas. Annu Rev Food Sci Technol 2011; 2:331–351.
4. Urashima T, Messer M, Oftedal O. Comparative biochemistry and evolution of milk oligosaccharides of monotremes, marsupials, and eutherians. In: Evolutionary biology: genome evolution, speciation, coevolution and origin of life. Switzerland: Springer International Publishing; 2014. pp. 3–33.
5. Mills S, Ross RP, Hill C, *et al.* Milk intelligence: mining milk for bioactive substances associated with human health. Int Dairy J 2011; 21:377–401.
6. Hickey RM. The role of oligosaccharides from human milk and other sources in prevention of pathogen adhesion. Int Dairy J 2012; 22:141–146.
7. ▪ Prado EL, Dewey KG. Nutrition and brain development in early life. Nutr Rev 2014; 72:267–284.

This article reviews the pathway from early nutrient deficiency to long-term brain function, cognition, and productivity.

8. ▪▪ Wright EM. Glucose transport families SLC5 and SLC50. Mol Aspects Med 2013; 34:183–196.

This article provides a thorough review on the molecular biology and biochemistry of the SGLT family, as well as its implications in disease.

9. Bisha I, Rodriguez A, Laio A, Magistrato A. Metadynamics simulations reveal a Na+ independent exiting path of galactose for the inward-facing conformation of vSGLT. PLoS Comput Biol 2014; 10:e1004017.
10. Augustin R. The protein family of glucose transport facilitators: it's not only about glucose after all. IUBMB Life 2010; 62:315–333.
11. Leturque A, Brot-Laroche E, Le Gall M. GLUT2 mutations, translocation, and receptor function in diet sugar managing. Am J Physiol Endocrinol Metab 2009; 296:E985–E992.
12. Roser M, Josic D, Kontou M, *et al.* Metabolism of galactose in the brain and liver of rats and its conversion into glutamate and other amino acids. J Neural Transm 2009; 116:131–139.
13. Sørensen M. Determination of hepatic galactose elimination capacity using 2-[18F] fluoro-2-deoxy-D-galactose PET/CT: reproducibility of the method and metabolic heterogeneity in a normal pig liver model. Scand J Gastroenterol 2011; 46:98–103.
14. Lange A, Grønbæk H, Vilstrup H, Keiding S. Age-dependency of galactose elimination capacity in healthy children and children with chronic liver disease. Scand J Gastroenterol 2011; 46:197–200.
15. Fridovich-Keil JL, Walter JH. Galactosemia. In: Valle D, Beaudet AL, Vogelstein B, *et al.*, editors. The online metabolic and molecular bases of inherited disease. New York: McGraw Hill; 2008. pp. 1–92.

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

16. Tang M, Wierenga K, Elsas LJ, Lai K. Molecular and biochemical characterization of human galactokinase and its small molecule inhibitors. Chem Biol Interact 2010; 188:376–385.

17. Chiappori F, Merelli I, Milanesi L, Marabotti A. Static and dynamic interactions between GALK enzyme and known inhibitors: guidelines to design new drugs for galactosemic patients. Eur J Med Chem 2013; 63:423–434.

18. McCorvie TJ, Timson DJ. The structural and molecular biology of type I galactosemia: enzymology of galactose 1-phosphate uridylyltransferase. IUBMB Life 2011; 63:694–700.

19. McCorvie TJ, Timson DJ. UDP-galactose-4-epimerase (GALE). In Taniguchi N, Honke K, Fukuda M, *et al.*, editors. Handbook of glycosyltransferases and related genes. Japan: Springer; 2014. pp. 1449–1464.

20. Décombaz J, Jentjens R, Ith M, *et al.* Fructose and galactose enhance postexercise human liver glycogen synthesis. Med Sci Sports Exerc 2011; 43:1964–1971.

21. Berry GT, Walter JH. Disorders of galactose metabolism. In Saudubray JM, van den Berghe G, Walter JH, editors. Inborn metabolic diseases: diagnosis and treatment. Germany: Springer; 2012.

22. Varki A, Esko JD, Colley KJ. Cellular organization of glycosylation. In: Varki A, Cummings RD, Esko JD, *et al.*, editors. Essentials of glycobiology. Cold Spring Harbor, New York: Cold Spring Harbor Laboratory Press; 2009.

23. ▪ Hadley B, Maggioni A, Ashikov A, *et al.* Structure and function of nucleotide sugar transporters: current progress. Comput Struct Biotechnol J 2014; 10:23–32.
This article provides an overview on the nucleotide sugar transporter family, focused on their structure–function relationship, and its implication in a number of diseases.

24. ▪▪ Cummings R, Pierce J. The challenge and promise of glycomics. Chem Biol 2014; 21:1–15.
This article provides an overview on glycans diversity and complexity, as well as their importance in human physiology and disease.

25. Berry GT. Is prenatal myo-inositol deficiency a mechanism of CNS injury in galactosemia? J Inherit Metab Dis 2011; 34:345–355.

26. Waisbren SE, Potter NL, Gordon CM, *et al.* The adult galactosemic phenotype. J Inherit Metab Dis 2012; 35:279–286.

27. Coss KP, Hawkes CP, Adamczyk B, *et al.* N-glycan abnormalities in children with galactosemia. J Proteome Res 2014; 13:385–394.

28. Coss KP, Treacy EP, Cotter EJ, *et al.* Systemic gene dysregulation in classical galactosaemia: is there a central mechanism? Mol Genet Metab 2014; 113:177–187.

29. Coss KP, Byrne JC, Coman DJ, *et al.* IgG N-glycans as potential biomarkers for determining galactose tolerance in Classical Galactosaemia. Mol Genet Metab 2012; 105:212–220.

30. Coelho AI, Trabuco M, Ramos R, *et al.* Functional and structural impact of the most prevalent missense mutations in classic galactosemia. Mol Genet Genom Med 2014; 2:484–496.

31. McCorvie TJ, Gleason TJ, Fridovich-Keil JL, Timson DJ. Misfolding of galactose 1-phosphate uridylyltransferase can result in type I galactosemia. Biochim Biophys Acta 2013; 1832:1279–1293.

32. Berry GT, Elsas LJ. Introduction to the Maastricht workshop: lessons from the past and new directions in galactosemia. J Inherit Metab Dis 2011; 34:249–255.

33. ▪ Salkovic-Petrisic M, Osmanovic-Barilar J, Knezovic A, *et al.* Long-term oral galactose treatment prevents cognitive deficits in male Wistar rats treated intracerebroventricularly with streptozotocin. Neuropharmacology 2014; 77:68–80.
This study reports a beneficial effect of oral galactose on learning and memory function in a nontransgenic rat model of sporadic Alzheimer's disease.

34. ▪▪ Tegtmeyer L, Rust S, van Scherpenzeel M, *et al.* Multiple phenotypes in phosphoglucomutase 1 deficiency. N Engl J Med 2014; 370:533–542.
This study demonstrated that phosphoglucomutase 1 deficiency is also a congenital disorder of glycosylation, in which galactose supplementation has proven beneficial.

1363-1950 Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.